

Roots & Returns

SCU Analytics Showdown



**GOOD
NATURE
AGRO**



Group 9 : Tanish

5254

FARMERS DID NOT SELL BACK





The Challenge

1

Predictive Model

Develop a model to predict expected farmer yield and identify the most significant predictors

2

Input Effectiveness

Identify the components of the loan package that are most strongly associated with improved yield

3

Risk Identification

Can high risk farmers be segmented and flagged early in the season

The 4 Datasets

Farmer Details

Gender, Age, Agro-ecological
Zone, Region, Number of Seasons

Loan Details

Input packages farmer received,
Any programmes involved in

Planting Survey

How many seeds planted,
Spacing used, When they were
planted

Buy Back Details

Who sold, How much they sold,
What price they received



01.

Predictive Model

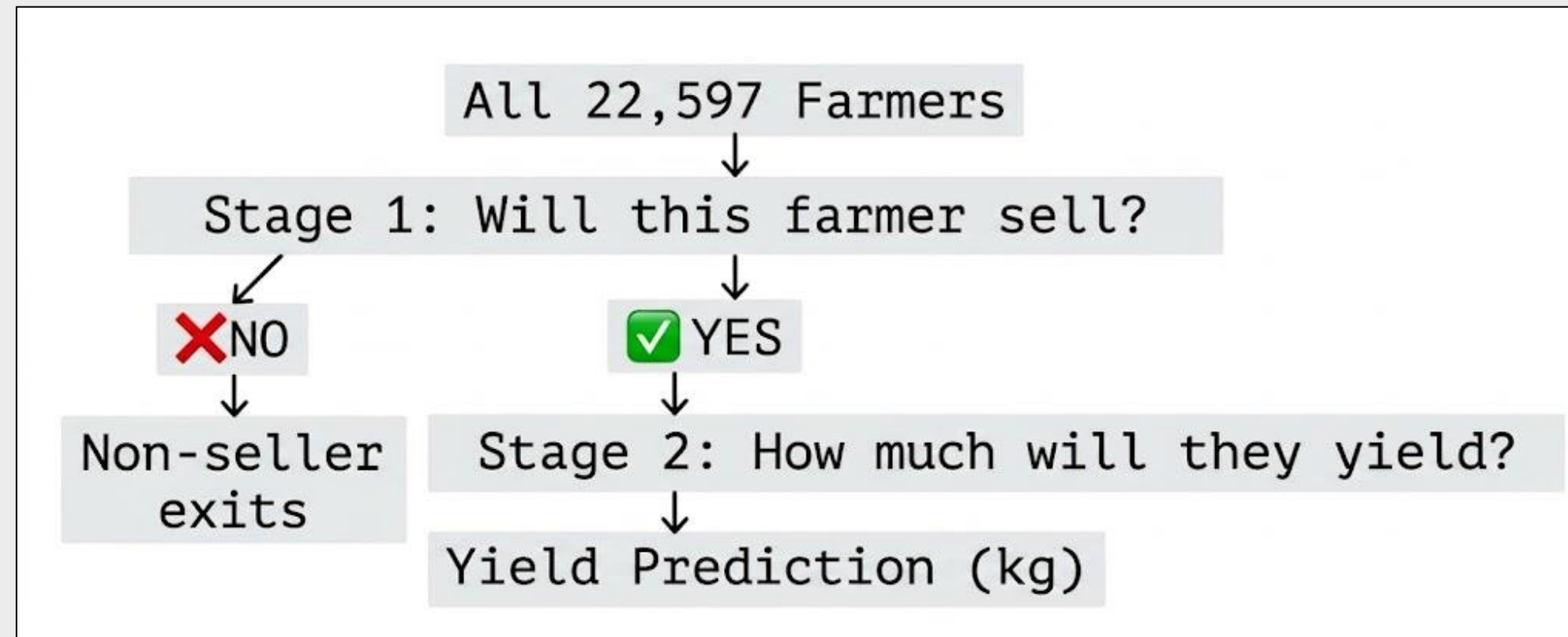
Develop a model to predict expected farmer yield and identify the most significant predictors

The Problem with Predicting Yield the Simple Way

- A standard regression model would fail, not because the model is poorly calibrated but because it is solving the wrong problem
- Ask a regressor to predict yield for a non-seller and it would return something like 80 kg — a small positive number
- 5254 Non-selling farmers didn't have a small harvest. They had no harvest.

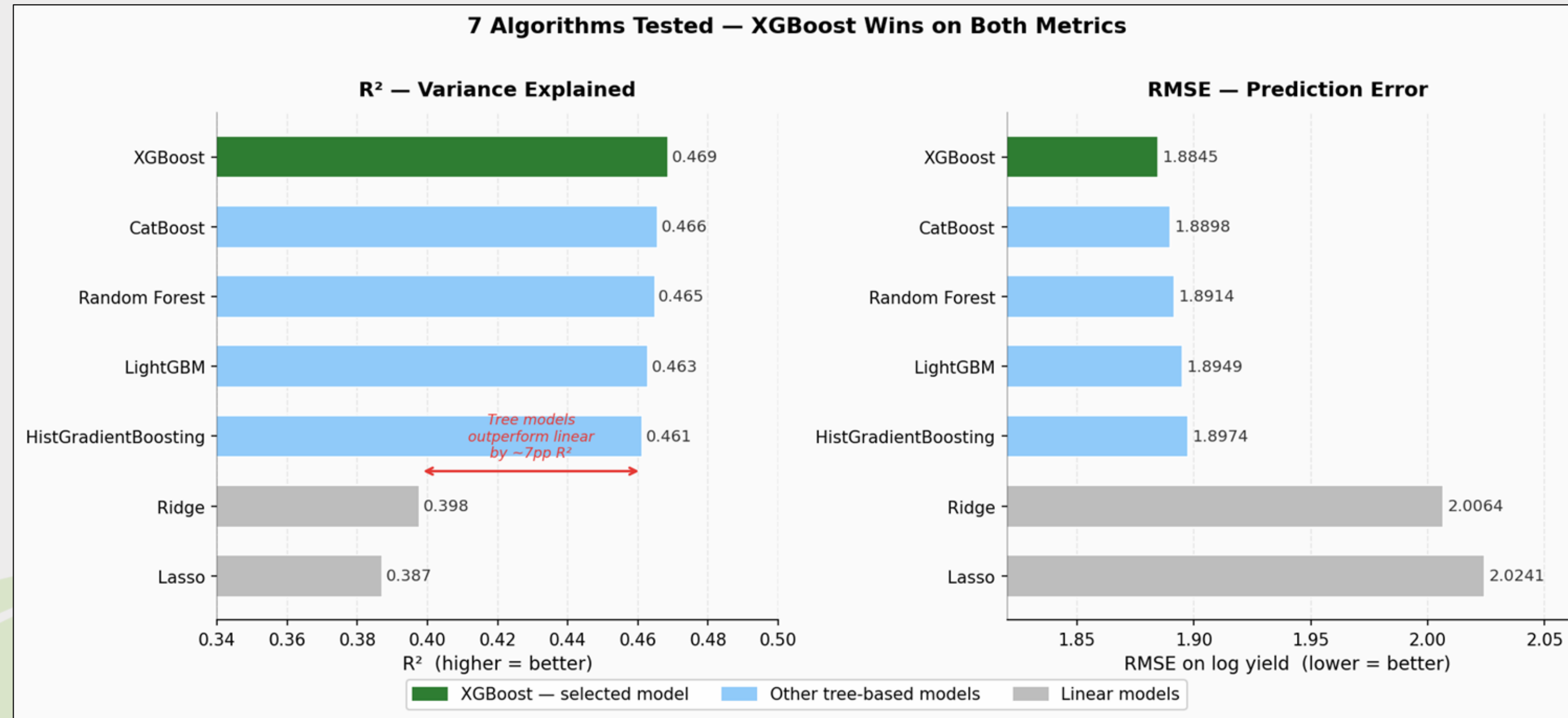
(Note: 2024-25 was a post-drought recovery season. Some non-sellers carry debt rolled over from the 2023-24 drought, making the non-seller problem especially acute this season.)

The Solution : Two Stage Hurdle Model



- **Stage 1 : Classifier**– Segregates farmer by seller vs. non seller
- **Stage 2 : Regressor** – If they sold, how much did they sell?
- The farmer must clear the first hurdle (the likelihood of selling) before the second stage makes any prediction about quantity

Picking the right model : XGBoost

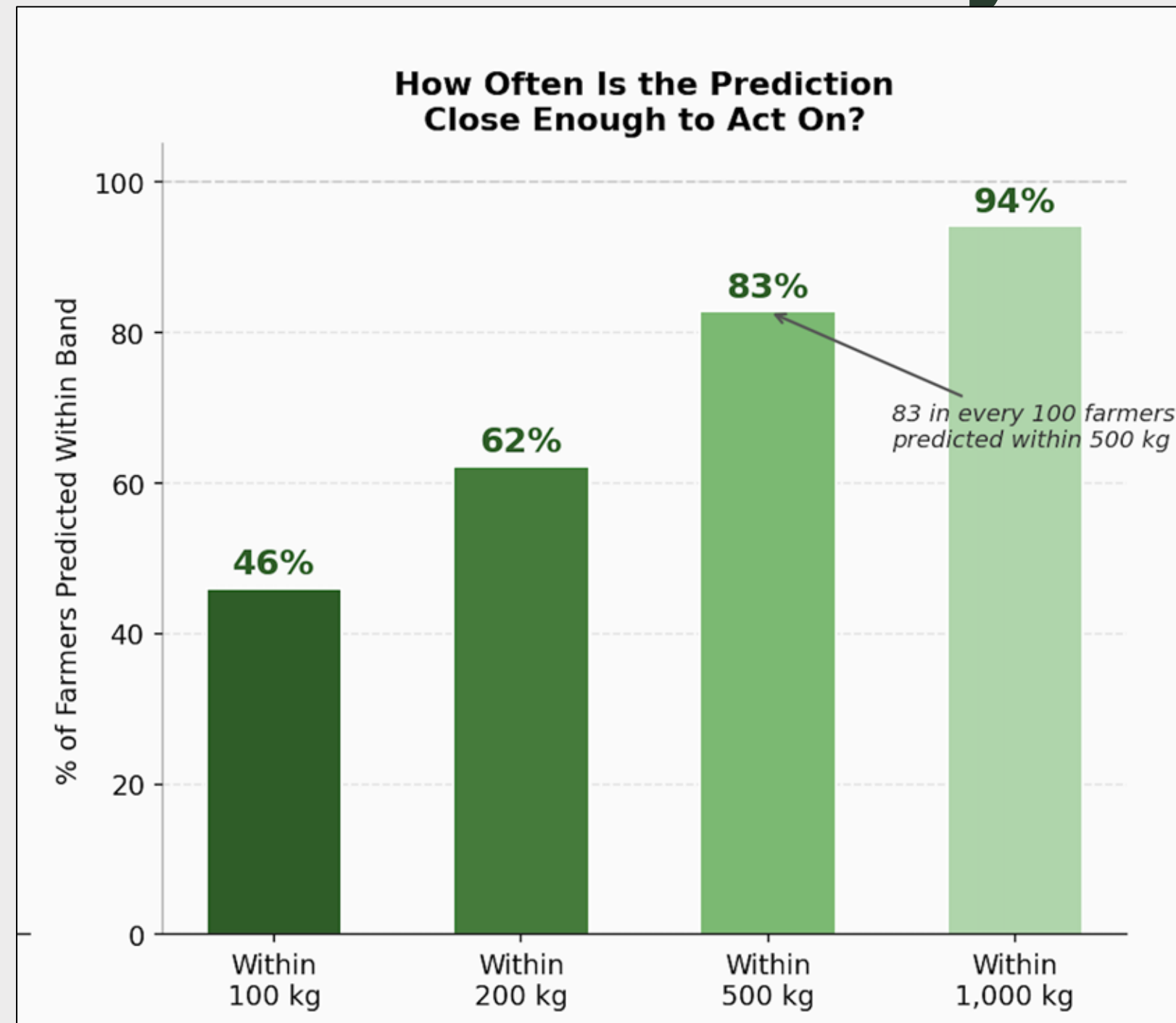


- 7 Algorithms tested on identical data and identical cross validation
- XGBoost won on every metric

1.8845
RMSE

0.469
R²

XGBoost Accuracy



- After selecting XGBoost, Bayesian hyperparameter tuning improved R^2 from 0.469 to 0.475
- The remaining variation is possibly due to **weather shocks (including drought recovery in 2023-24), soil micro-conditions, and unobserved farmer decisions** — factors no model can see



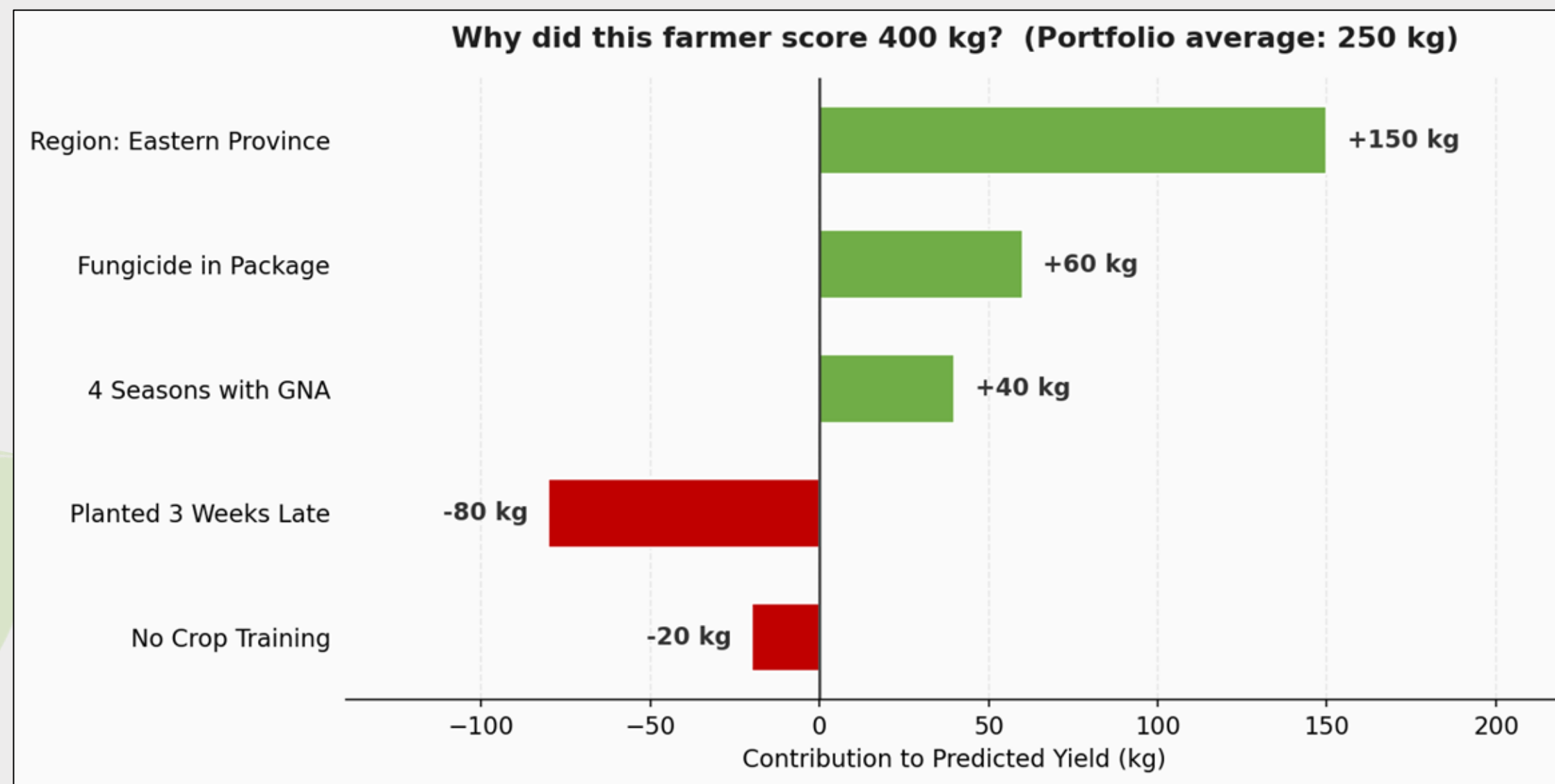
SHapley Additive exPlanations gives us
not just a prediction as to what this
farmer will yield, it tells us:

"Why does the model predict it and
which specific factors are driving it?"

Understanding SHAP

An Example of how SHAP works

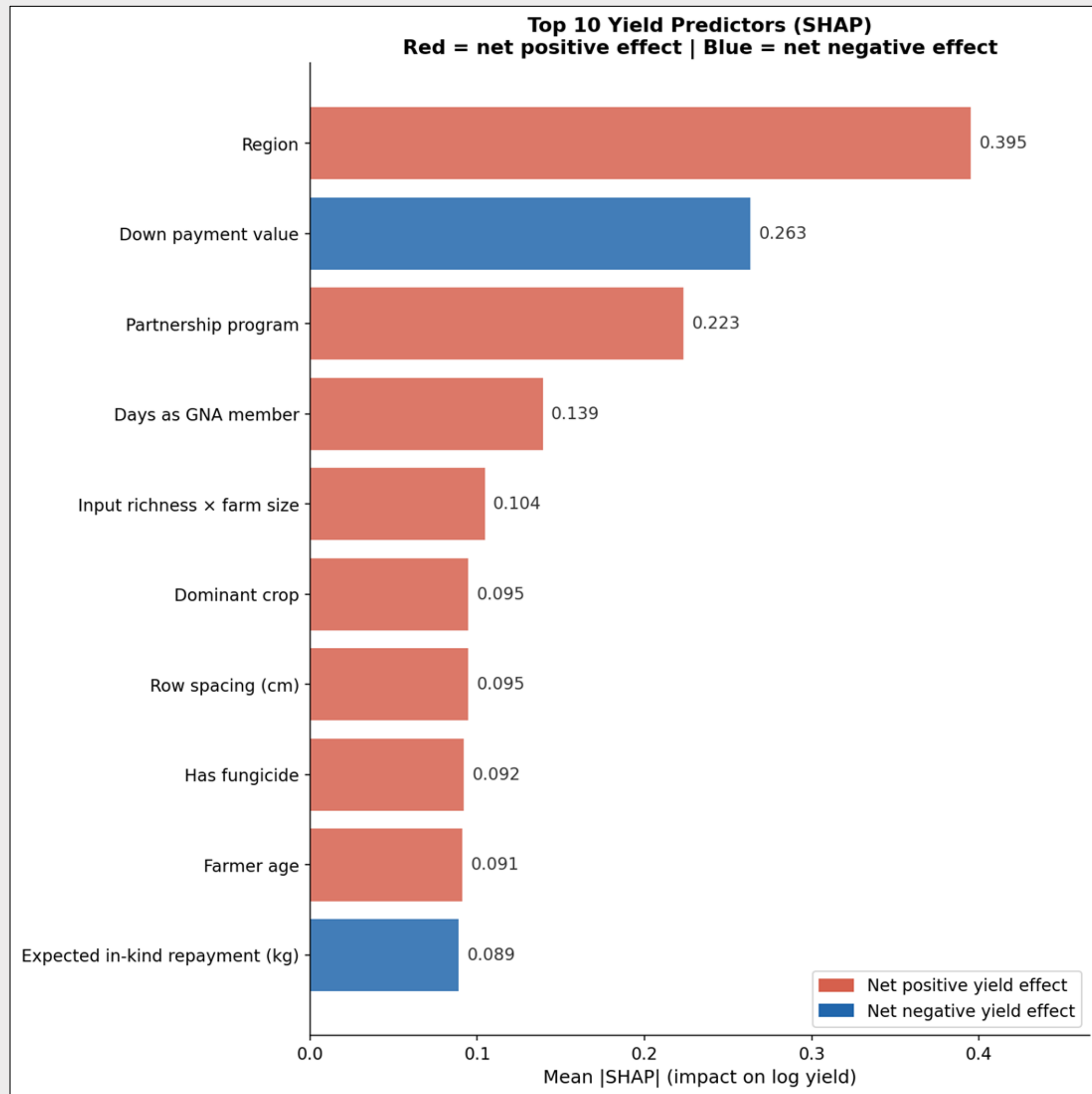
- If our model says “a farmer will yield 400 kg”. Good Nature Agro needs more than a number — they need to know which factors drove it and which inputs genuinely caused better outcomes!



Every Factor Gets a Score and is Ranked. Positive = Pushed Yield Up. Negative = Pulled Yield Down.

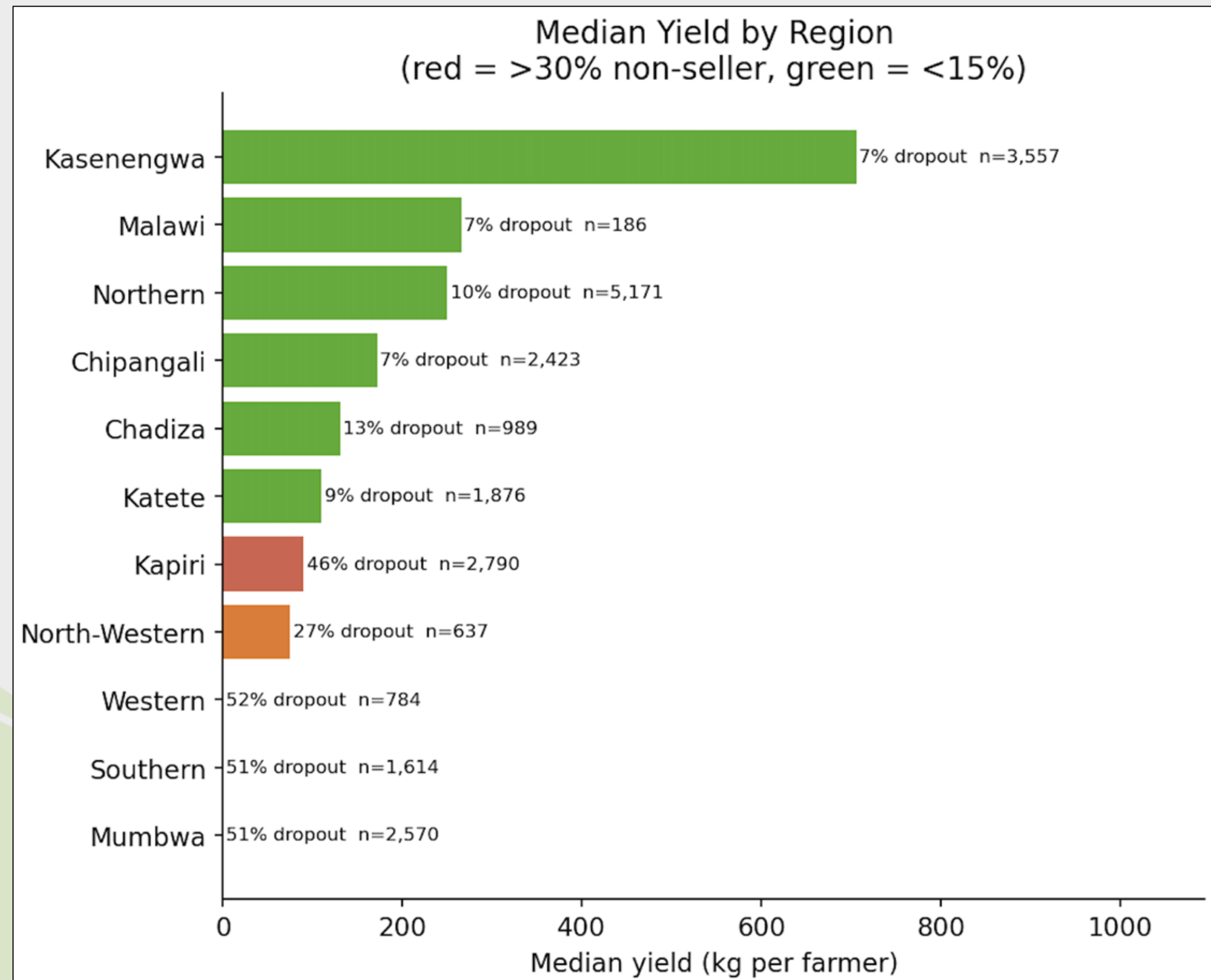
True Drivers of Yield

Applying SHAP for our 22,597 Farmers
A Clear Picture Emerges.



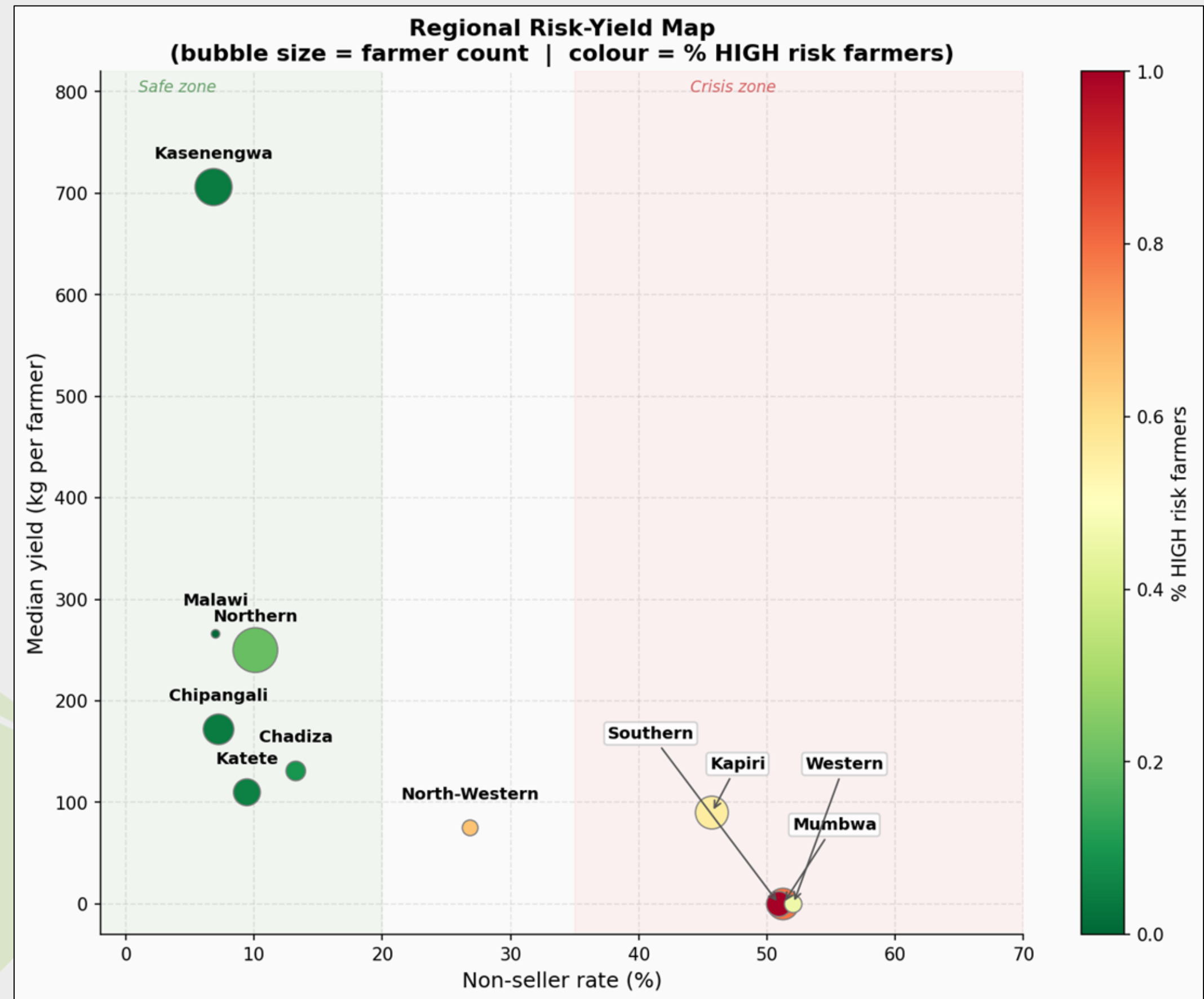
Insight 1 : Geography is Destiny

- **Kasenengwa leads the portfolio – 706 kg median yield, only 7% dropout across 3,557 farmers**
- **Three regions have collapsed – Western, Southern and Mumbwa each sit above 50% dropout with near-zero median yield. Combined, that is over 4,900 farmers delivering almost nothing**
- **Kapiri is the warning sign – 2,790 farmers, 46% dropout, still showing some yield. This is a region mid-collapse, not yet at zero but heading there**
- **Low dropout and high yield move together – every green region has both. Every red region has neither. Geography determines both outcomes simultaneously**



Insight 1 : Geography is Destiny

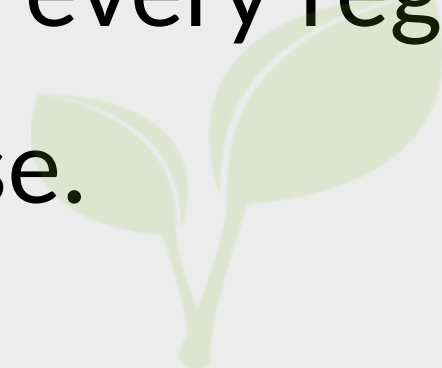
- **Safe Zone** — Kasenengwa, Northern, Chipangali, Katete, Chadiza and Malawi : low non-seller rates, meaningful yield, deep green. These regions are working
- **Crisis Zone** — Western, Southern and Mumbwa (regions likely disproportionately affected by the 2023-24 drought) : 50%+ non-seller rates, near-zero yield, deep red. These are highly underperforming zones
- Northern is the most important region to protect — largest bubble in the safe zone, 5,171 farmers, 10% dropout. Any deterioration here has the largest impact



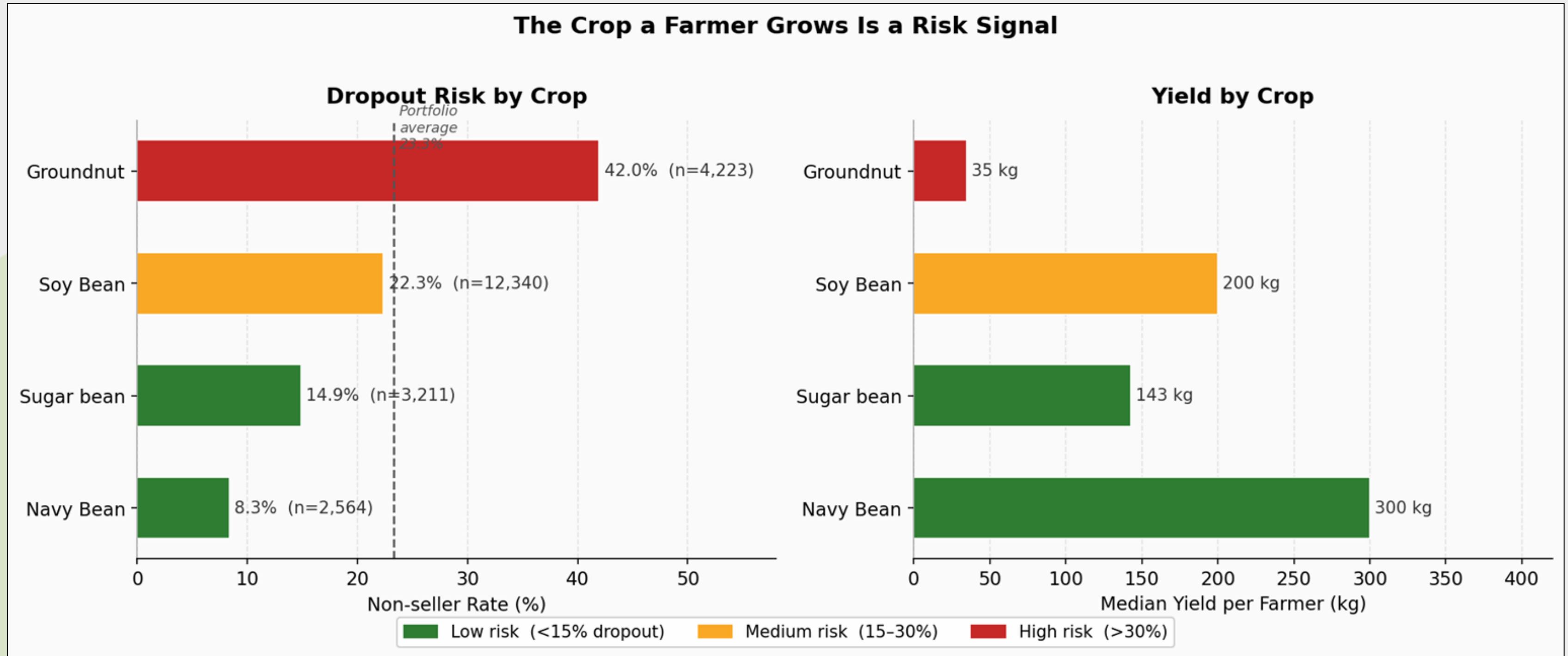


Business Insight

Non-seller rates are geographically structured — not random. This means interventions can be geographically targeted. Not every region needs the same response.



Insight 2 : The Crop a Farmer Grows is a Risk Signal

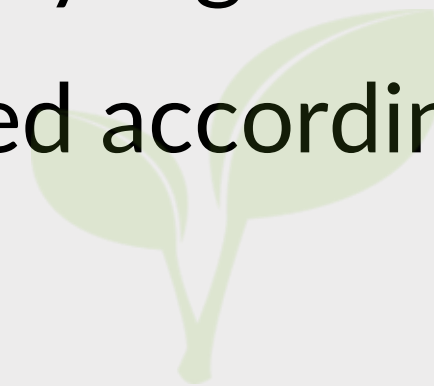


- **Groundnut:** 42% non-seller rate | Median yield 35 kg
- **Soy Bean:** 22% non-seller rate | Median yield 200 kg
- **Navy Bean:** 8.3% non-seller rate | Median yield 300 kg



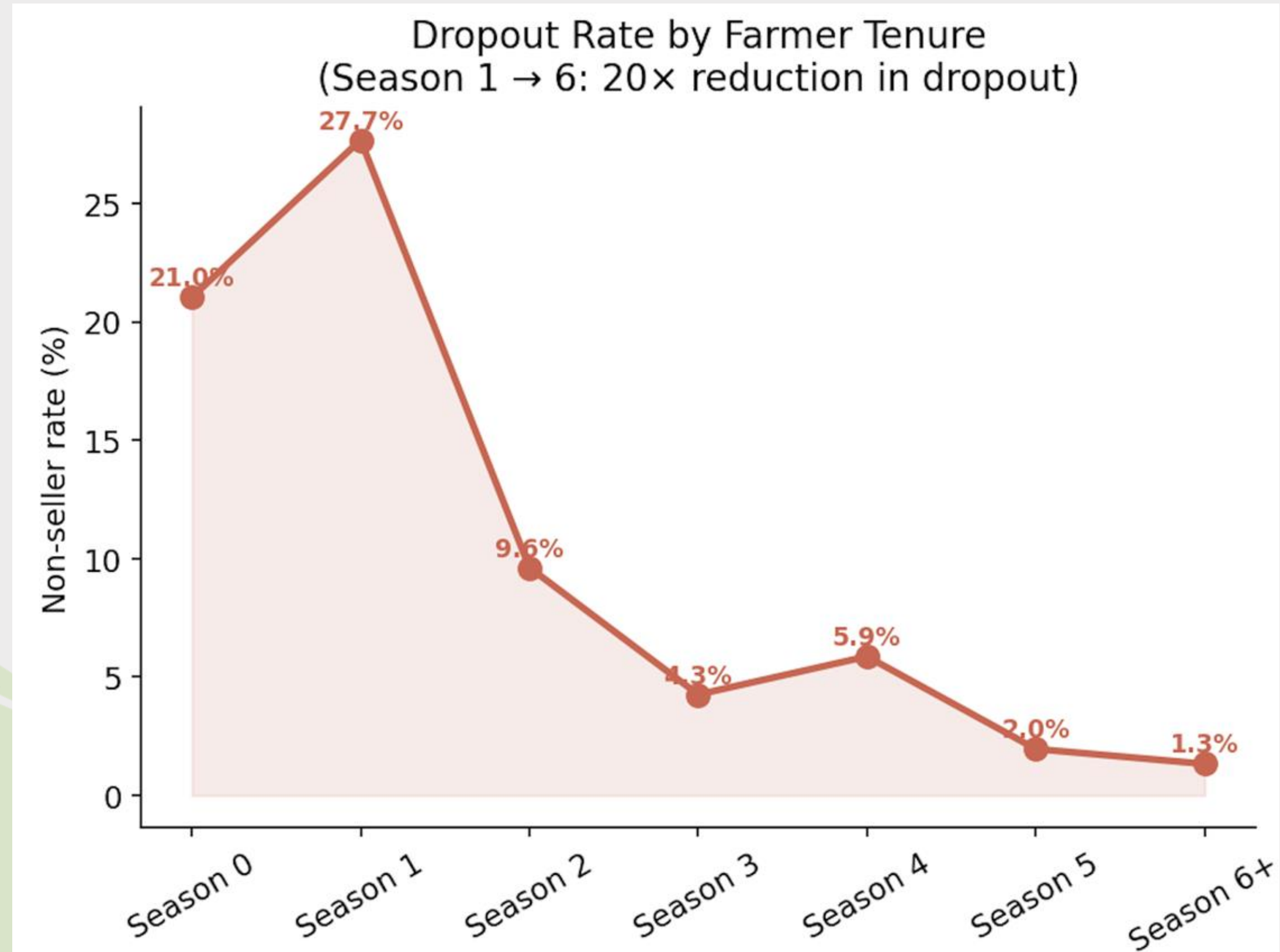
Business Insight

Crop selection at loan registration is not just an agronomic decision, it is a risk decision. Certain crop-zone combinations carry structurally higher dropout risk and should be priced accordingly.



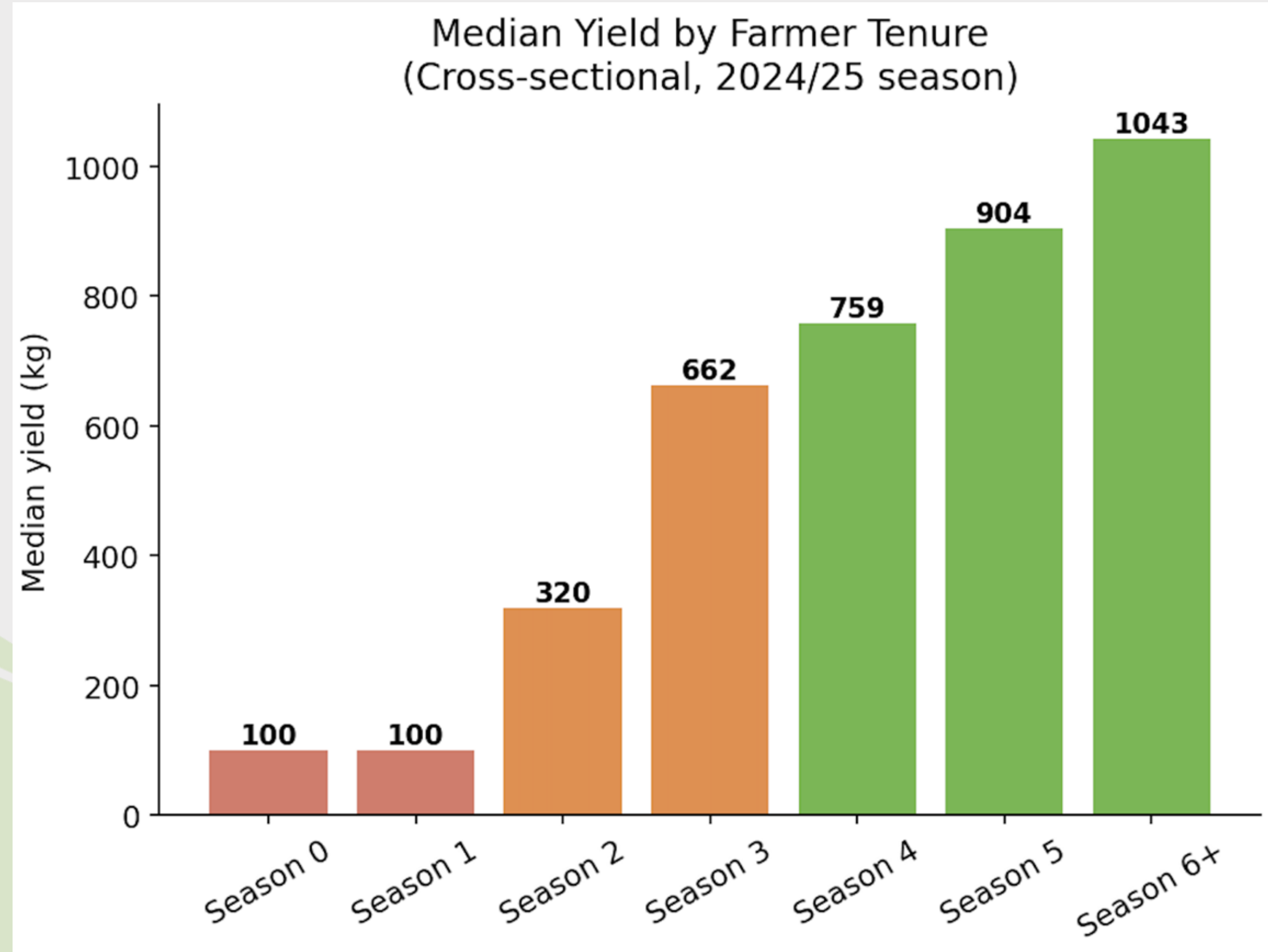
Insight 3 : Loyalty is Key

- **Season 1 is the danger point** — dropout peaks at 27.7%, the highest of any season. This is where Good Nature Agro loses the most farmers
- Survive Season 1 and the dropout falls from **27.7%** to **9.6%** by Season 2, then to **4.3%** by Season 3



Insight 3 : Loyalty is Key

- Yield follows the same curve — **100 kg** at Season 1, **320 kg** by Season 2, **1,043 kg** by Season 6+. **A 10x increase over the farmer's lifetime with Good Nature Agro**
- Season 6+ farmers are Good Nature Agro's most reliable asset with **1.3%** dropout rate, **1,000+ kg** median yield. These farmers have minimal churn risk
- **The first two seasons are make or break** — every farmer who survives Season 1 and 2 is on a trajectory toward long-term productivity and loyalty





Business Insight

Invest disproportionately in Season 1 farmer support. It is the single intervention with the highest long-term return in Good Nature Agro's entire portfolio.





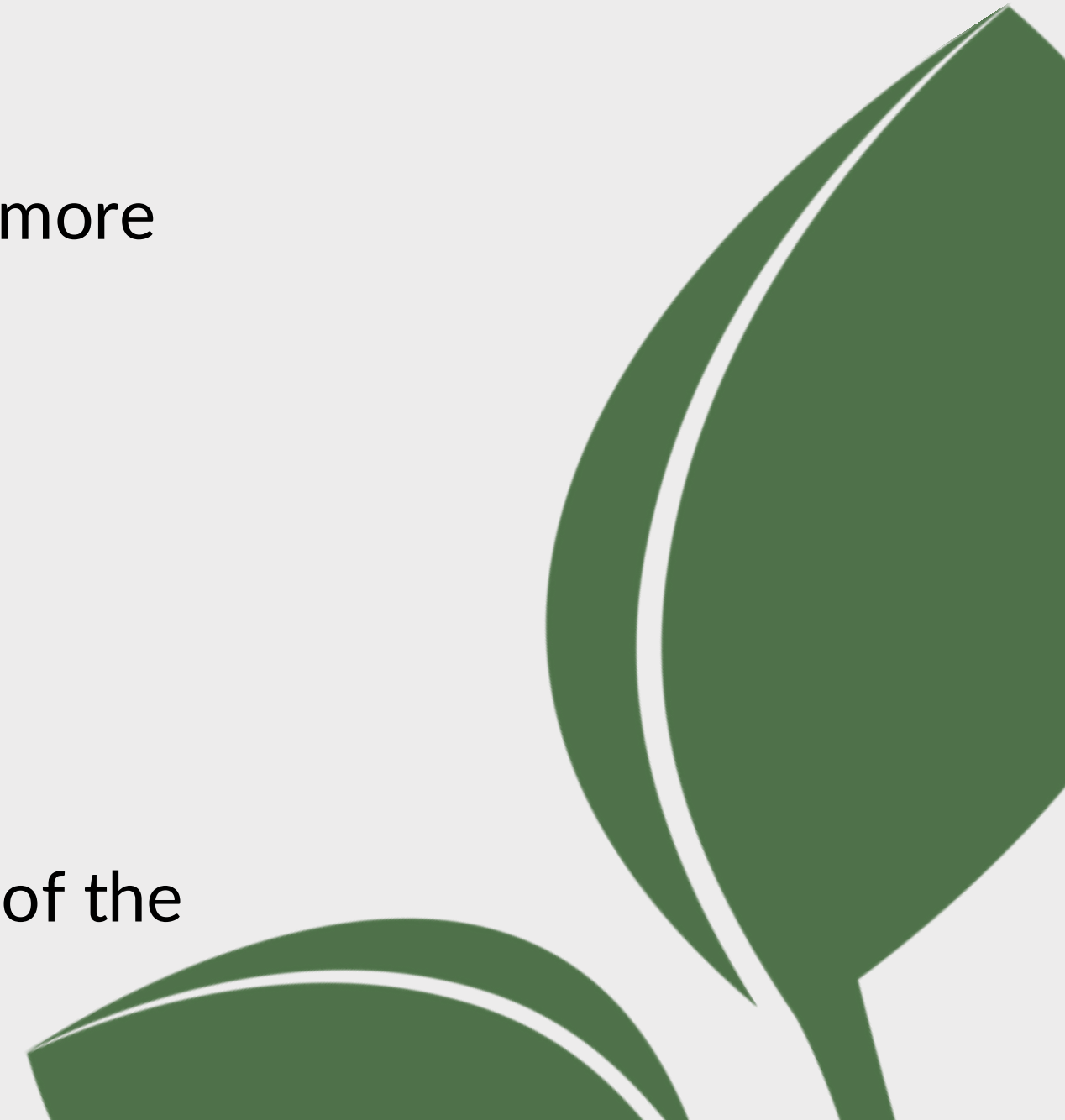
02.

Input Effectiveness

Identify the components of the loan package that are most strongly associated with improved yield

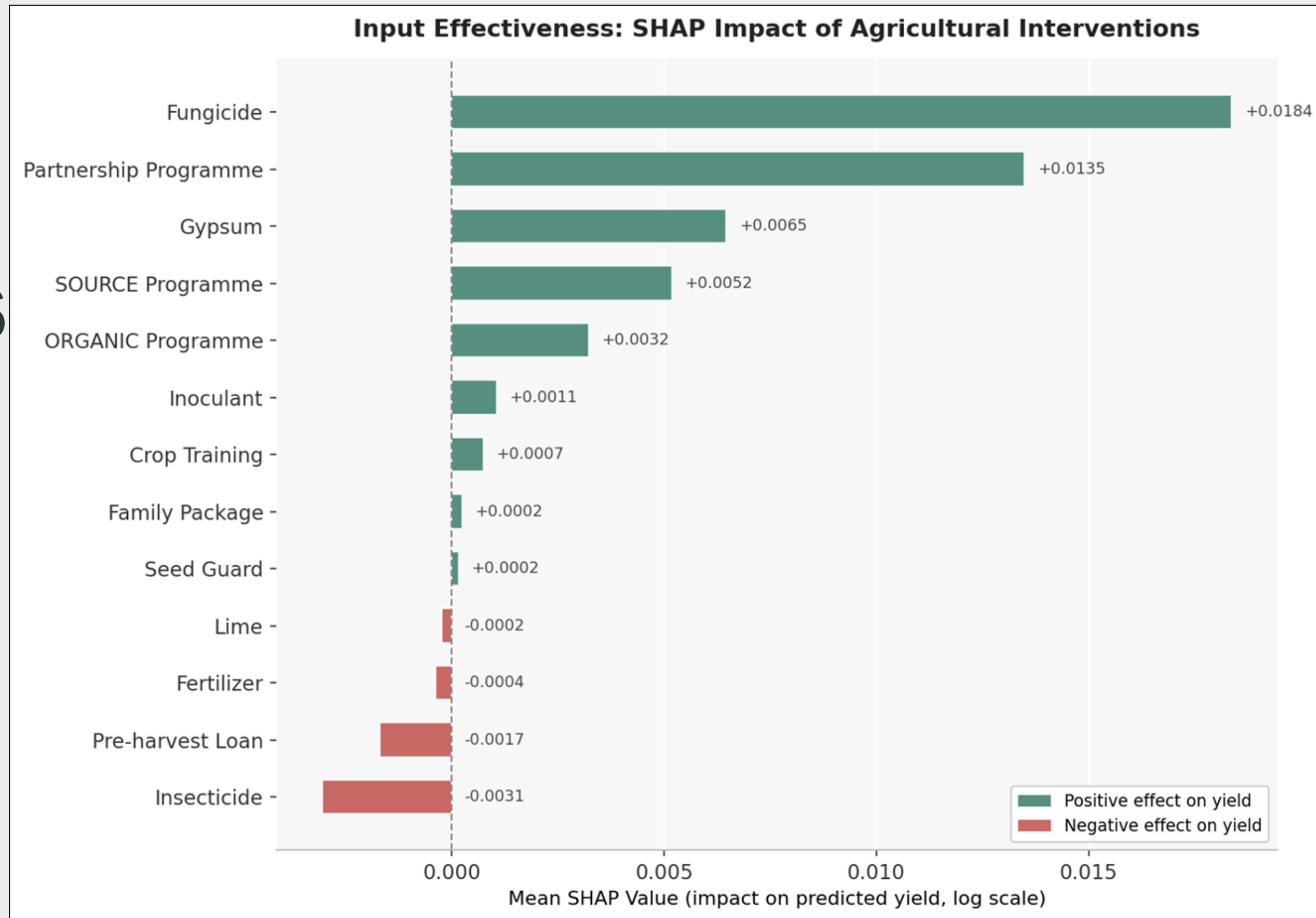
Which Inputs Genuinely Drive Yield?

- Good Nature Agro provides fertilizer, fungicide, gypsum, lime, inoculant, insecticide and seed guard
- **The problem:** Farmers who use more inputs may also tend to be more experienced, in better zones, or on larger farms
- If we just compare "used input" vs "didn't use input" — we are comparing different types of farmers, not the input itself
- **SHAP** controls for all of this. What remains is the genuine effect of the input alone.



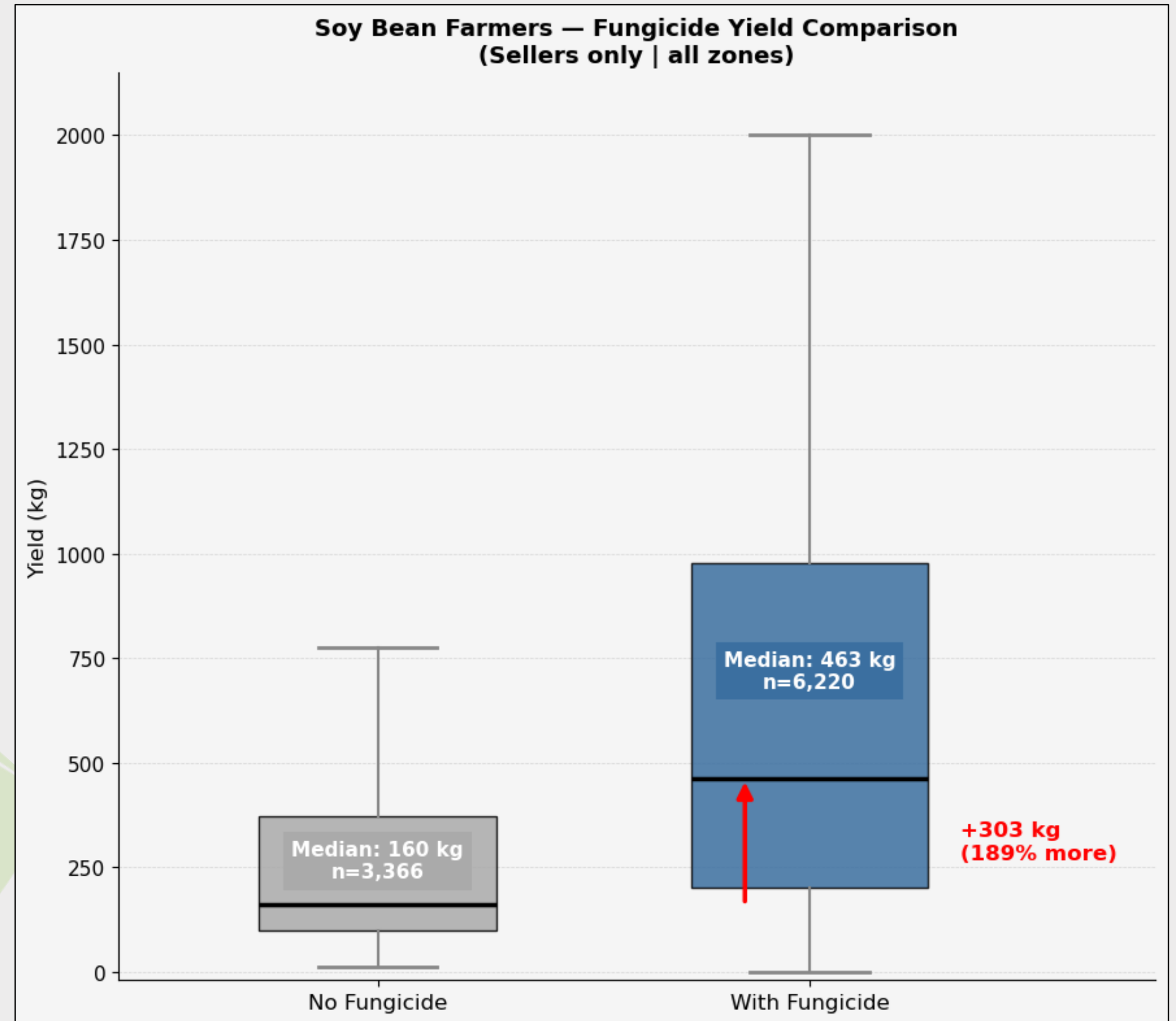
Impact of Inputs (Interventions)

Applying SHAP for our 22,597 Farmers
A Clear Picture Emerges.



Insight 4 : The Highest-Impact Input that most Soy Bean Farmers Don't Have

- **Fungicide** ranks as the #1 controllable input in the entire model with a consistent positive effect across crop types
- Soy Bean is the largest crop type at 55% of the portfolio
- Among Soy Bean farmers across all zones — those **with fungicide yield a median of 463 kg**
- Those **without fungicide yield a median of 160 kg** — the same crop, the same zones, a **189%** difference
- Yet, **45%** of Soy Bean farmers currently have no fungicide in their loan package



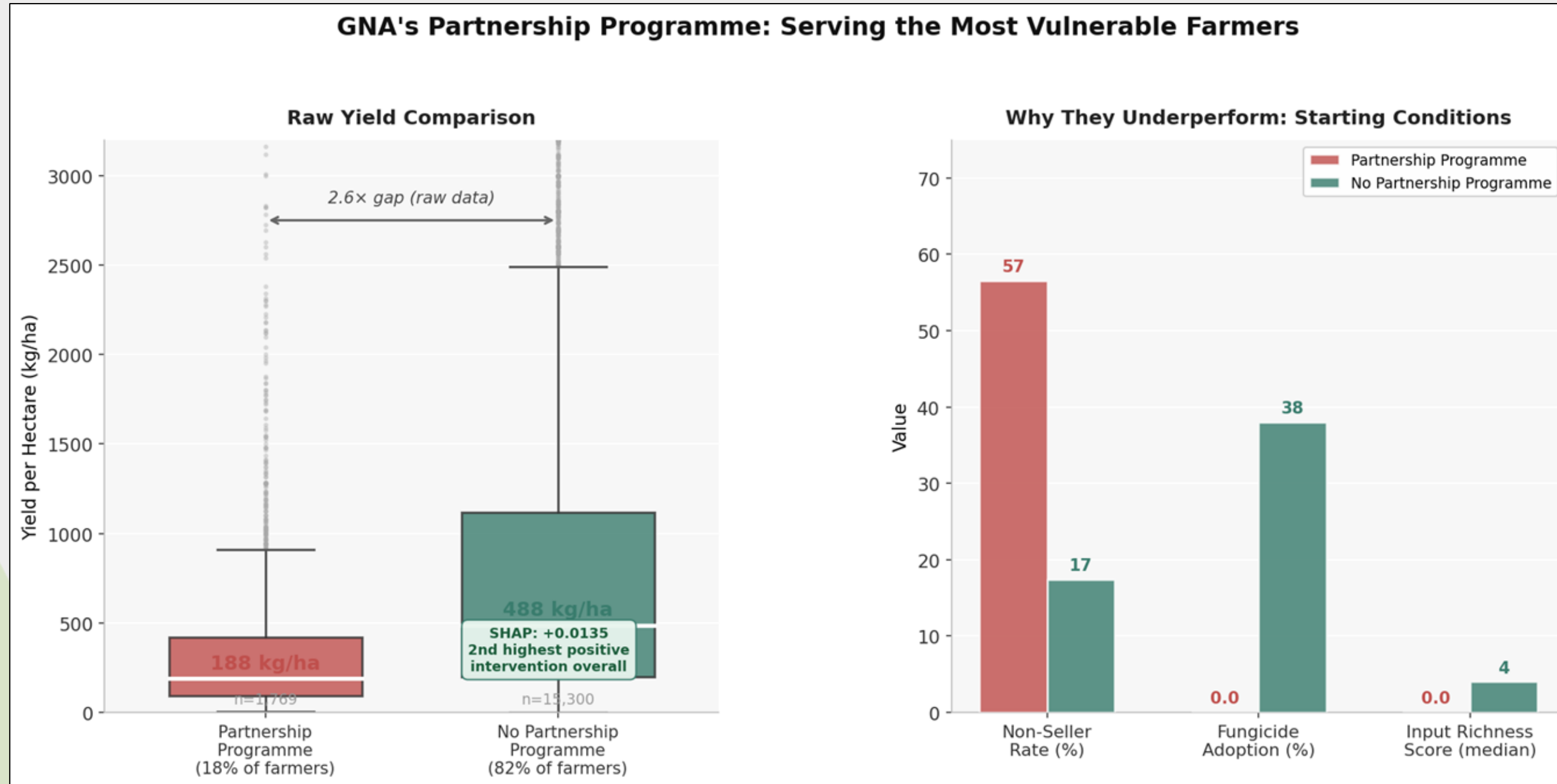


Business Insight

Making fungicide the default in every Soy Bean package is the highest-return input decision Good Nature Agro can make.



Insight 5 : The Partnership Programme Is Working – Just Not Where You'd Expect



- 18% of farmers are enrolled in the Partnership Programme – concentrated in Mumbwa, Kapiri, and North-Western
- On the surface: programme farmers yield 188 kg/ha vs 488 kg/ha for others, a 2.6 times gap
- **But this is misleading:** programme farmers have 0% fungicide adoption, input richness score of 0, and a 57% non-seller rate and they belong to Good Nature Agro's most disadvantaged cohort
- After controlling for all these disadvantages, the model assigns the programme a **SHAP** of +0.0135 – the 2nd highest positive intervention in the entire dataset
- Hence, the programme isn't failing, it's serving the hardest cases and still adding measurable value

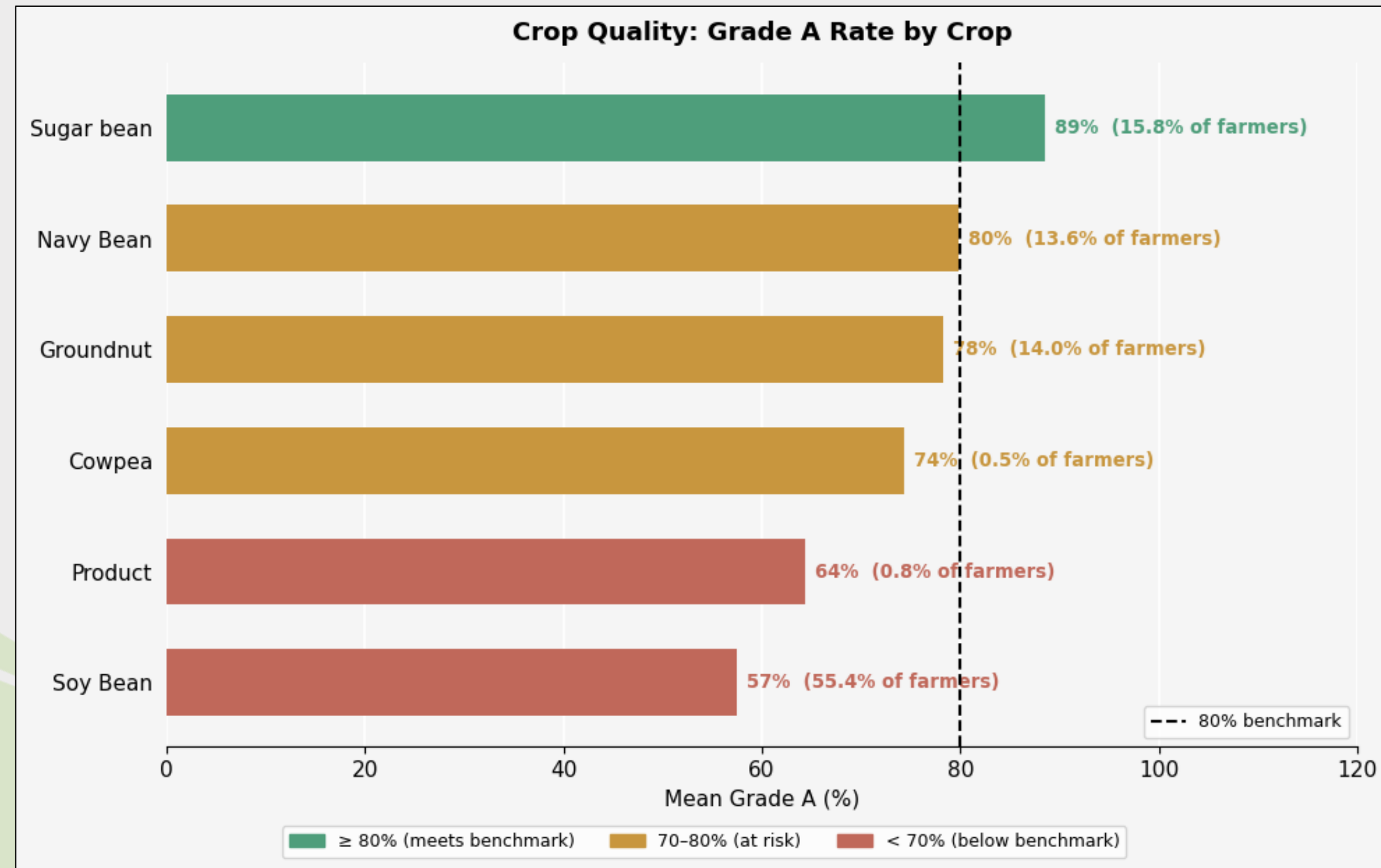


Business Insight

The Partnership Programme is Good Nature Agro's most important safety net — pairing it with even one additional input (fungicide or inoculant) for enrolled farmers is the highest-leverage investment to make.

Insight 6 : The Poor Quality of Soy Bean

- Soy Bean accounts for **55.4%** of all farmers — yet has the **lowest Grade A rate** of any major crop at **57%**, well below the 80% benchmark
- Sugar Bean is the only crop comfortably above benchmark at **89%**
- Navy Bean sits exactly at the **80%** benchmark and might be at risk of falling below the benchmark





Business Insight

Over half of Good Nature Agro's farmer base is producing below-benchmark quality. Soy Bean is the largest part of the portfolio, yet suffers from the poorest quality. Targeted agronomic support for Soy Bean farmers is the highest-impact quality intervention available.



03.

Risk Identification

Can high risk farmers be segmented and flagged early in the season

How Risk Tiers are assigned

- Each farmer gets a **single score** (*Expected Yield*) — that combines both **the chance they sell back** **AND** **how much they'll produce if they sell back**
- The Formula :
$$\text{Expected Yield} = \underset{\substack{\text{Stage 1} \\ \text{(classifier)}}}{P(\text{sells back})} \times \underset{\substack{\text{Stage 2} \\ \text{(regressor)}}}{E[\text{yield} \mid \text{sells back}]}$$
- We segment farmers into equal thirds by expected yield — the bottom third flagged **HIGH**, middle third **MEDIUM**, top third **LOW**.



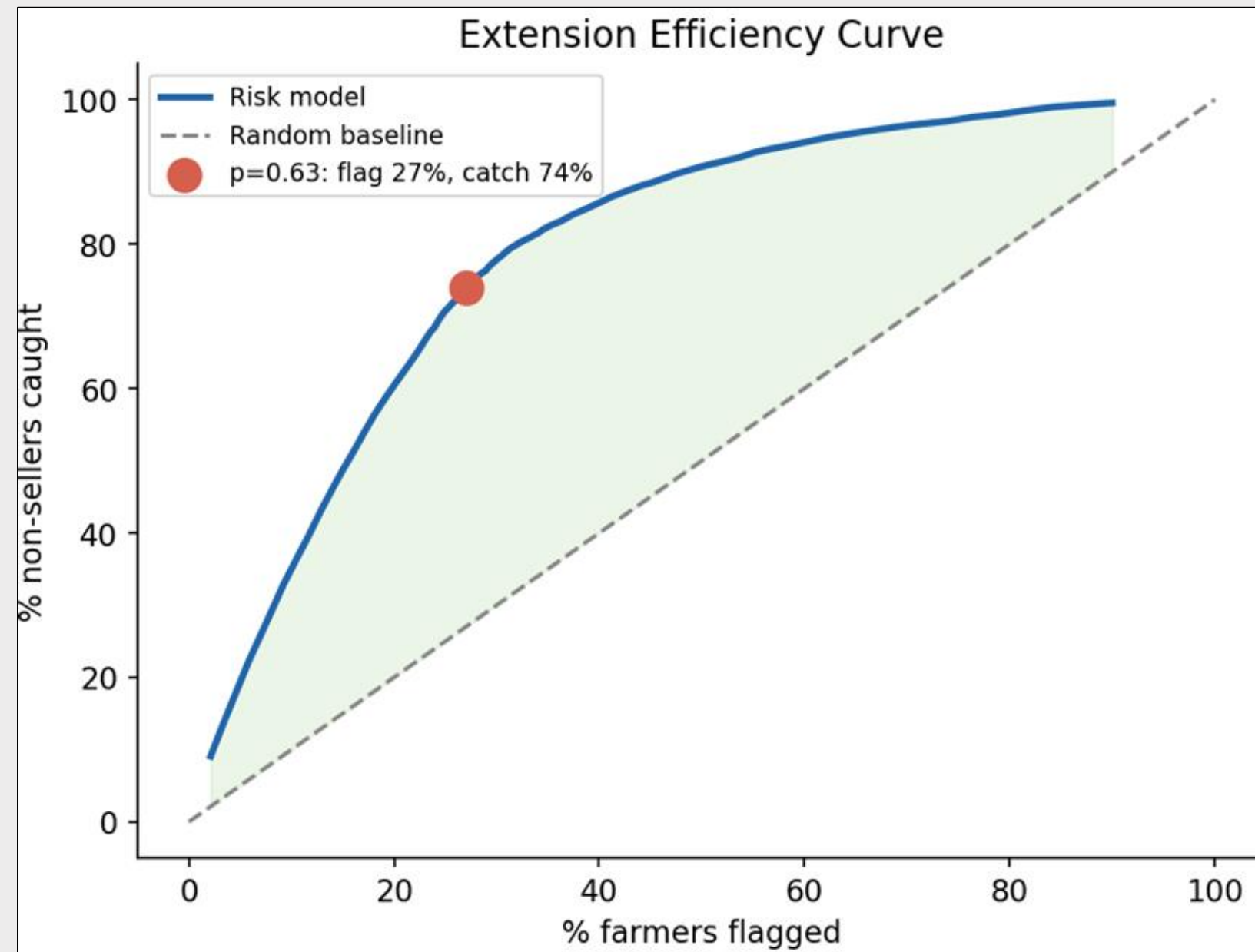
Risk Tier Thresholds

Tier	Expected Yield	Avg P(sell back)	Actual Non-Seller Rate
● HIGH	≤ 83 kg	26%	54.5%
● MEDIUM	84 – 216 kg	75%	11.7%
● LOW	> 216 kg	89%	3.9%

*(The 83 kg and 216 kg cutoffs emerge directly from the dataset.

- Thresholds are set at the **33rd and 67th percentiles** — equal thirds of the farmer population
- A farmer in HIGH tier has only a **26%** chance of selling back & their expected contribution is nearly zero
- This single score collapses two separate model outputs into one actionable, field-ready label

Tier Model Accuracy : The Model catches 3 in 4 non-sellers!



- By flagging only **27%** of the farmers, the model **catches 74% of non-sellers**
- A random approach would require flagging 74% of farmers to achieve the same catch rate — **2.7× less efficient**
- We can reach 3 out of 4 non-sellers while spending roughly a quarter of the extension costs

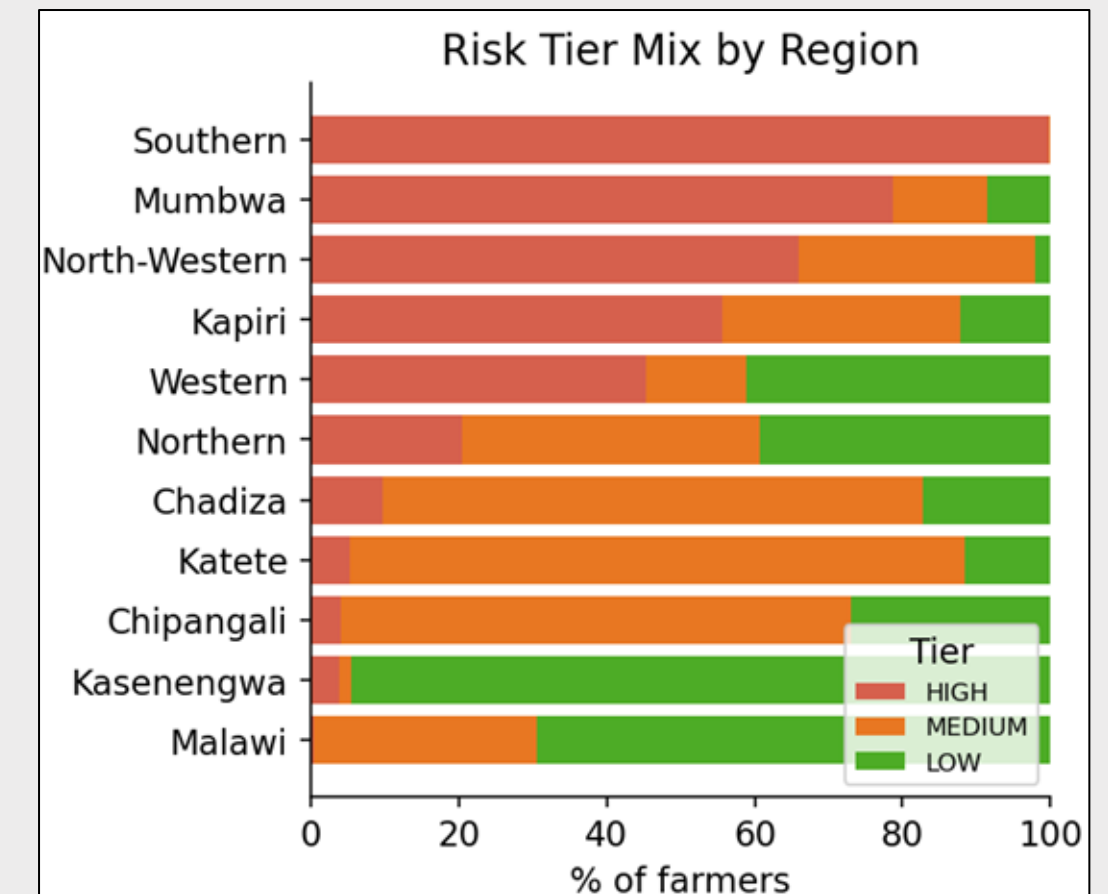
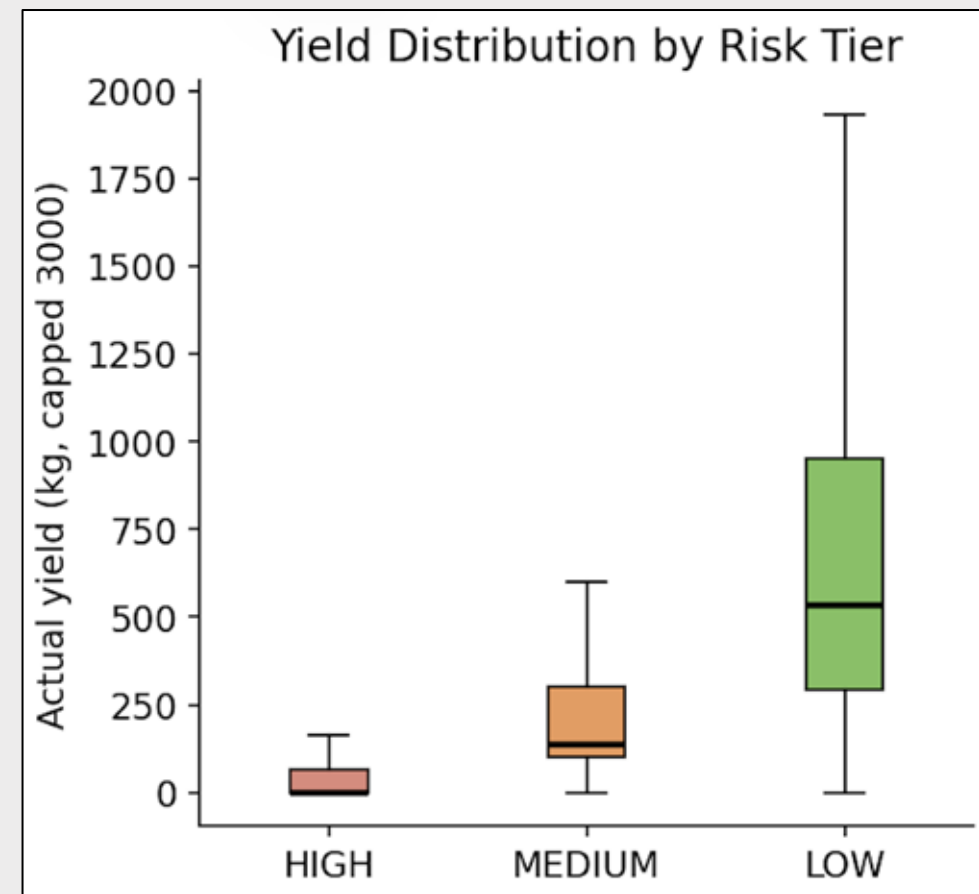
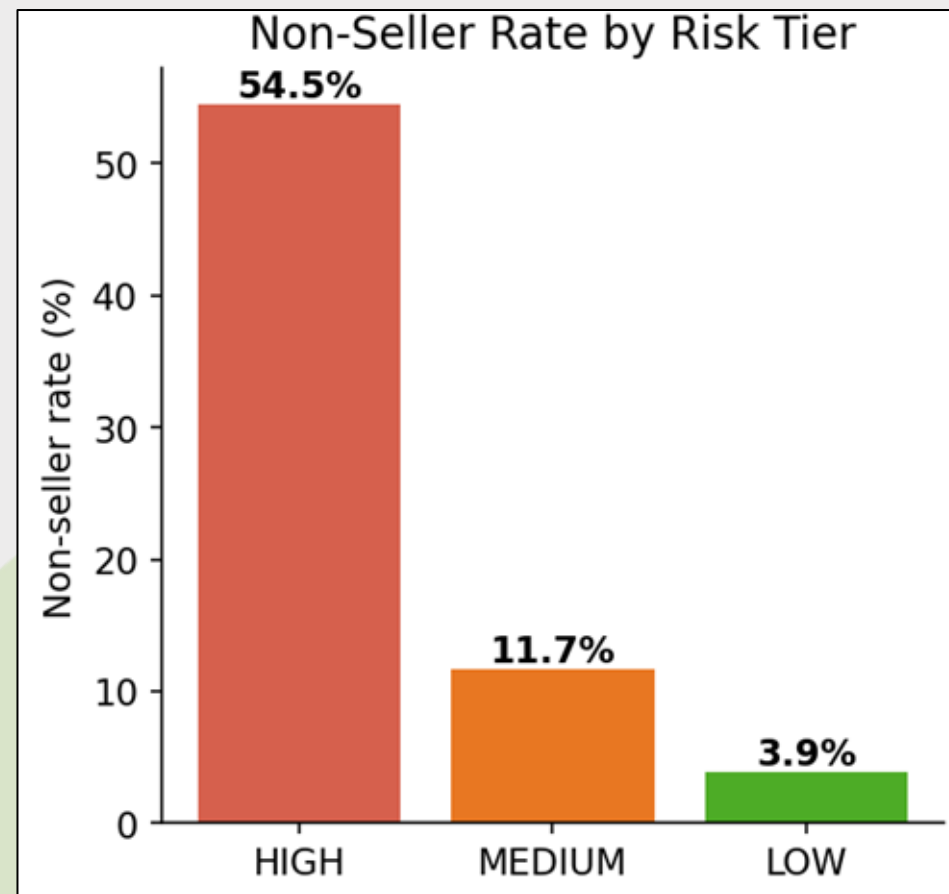


Business Insight

By leveraging this risk tier model, Good Nature Agro can implement targeted field visits, control loan exposure, and execute early interventions for HIGH risk farmers

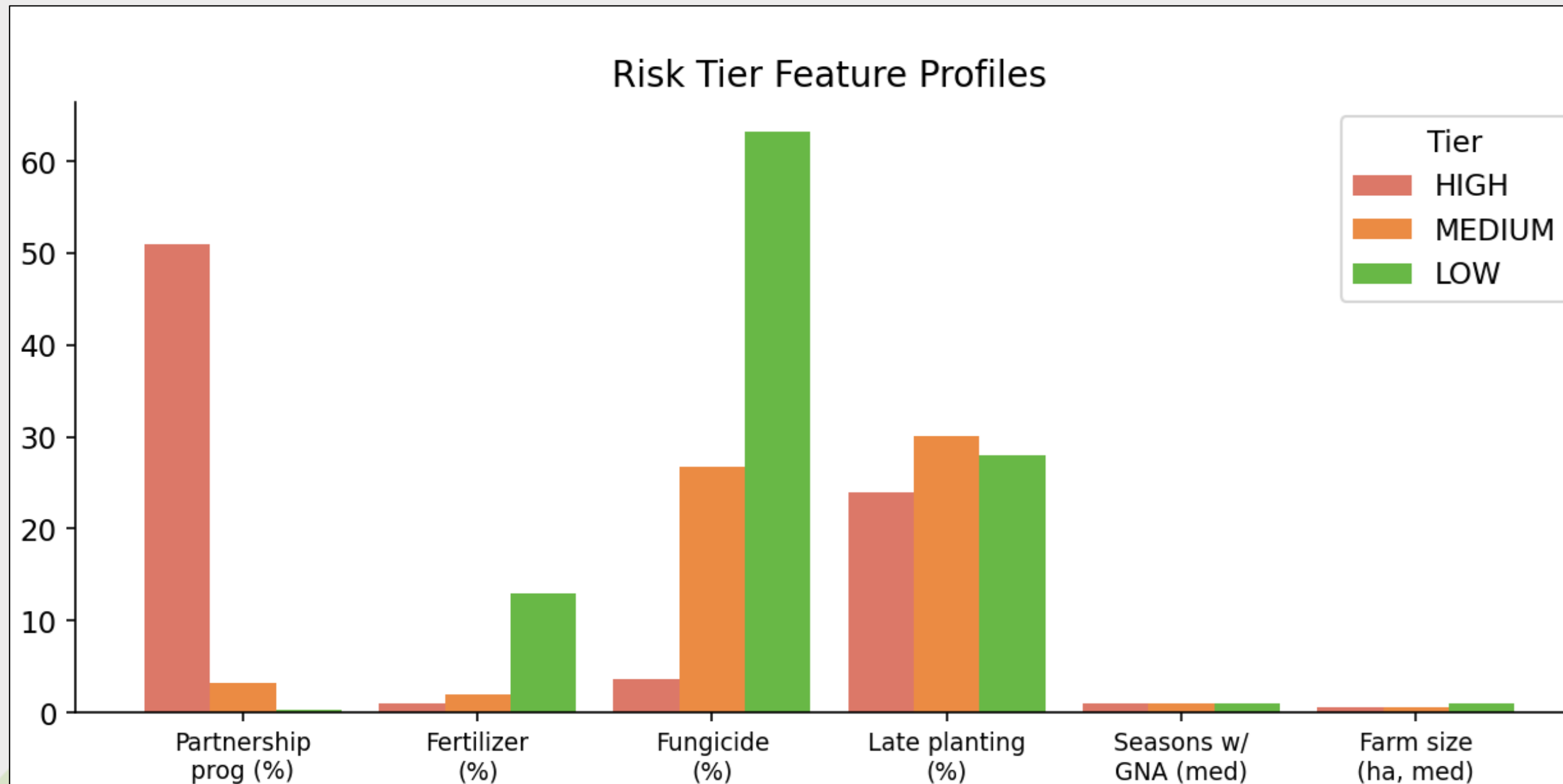


Insight 7 : 54% of HIGH-Risk Farmers Never Sell Back!



- **Non-seller rate: HIGH = 54.5% | MEDIUM = 11.7% | LOW = 3.9%** — the model cleanly separates farmers who will and won't sell back
- Yield follows the same pattern: **HIGH-tier median yield is near zero; LOW-tier median is ~550 kg**
- **Risk is regionally related:** Southern and Mumbwa are almost entirely HIGH risk; Kasenengwa is predominantly LOW risk

Insight 8 : The Profile of a High-Risk Farmer

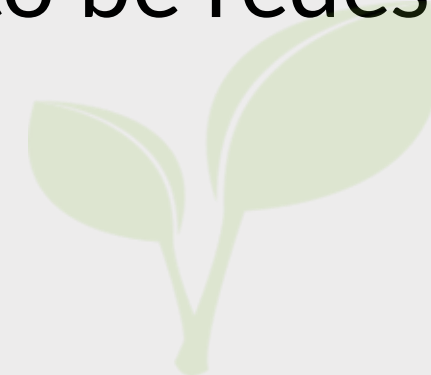


- **Partnership Programme** farmers make up **51%** of HIGH Tier vs just **2%** of LOW Tier
- **Fungicide** adoption is **4%** in HIGH Tier vs **63%** in LOW Tier, suggesting fungicide drives low risk

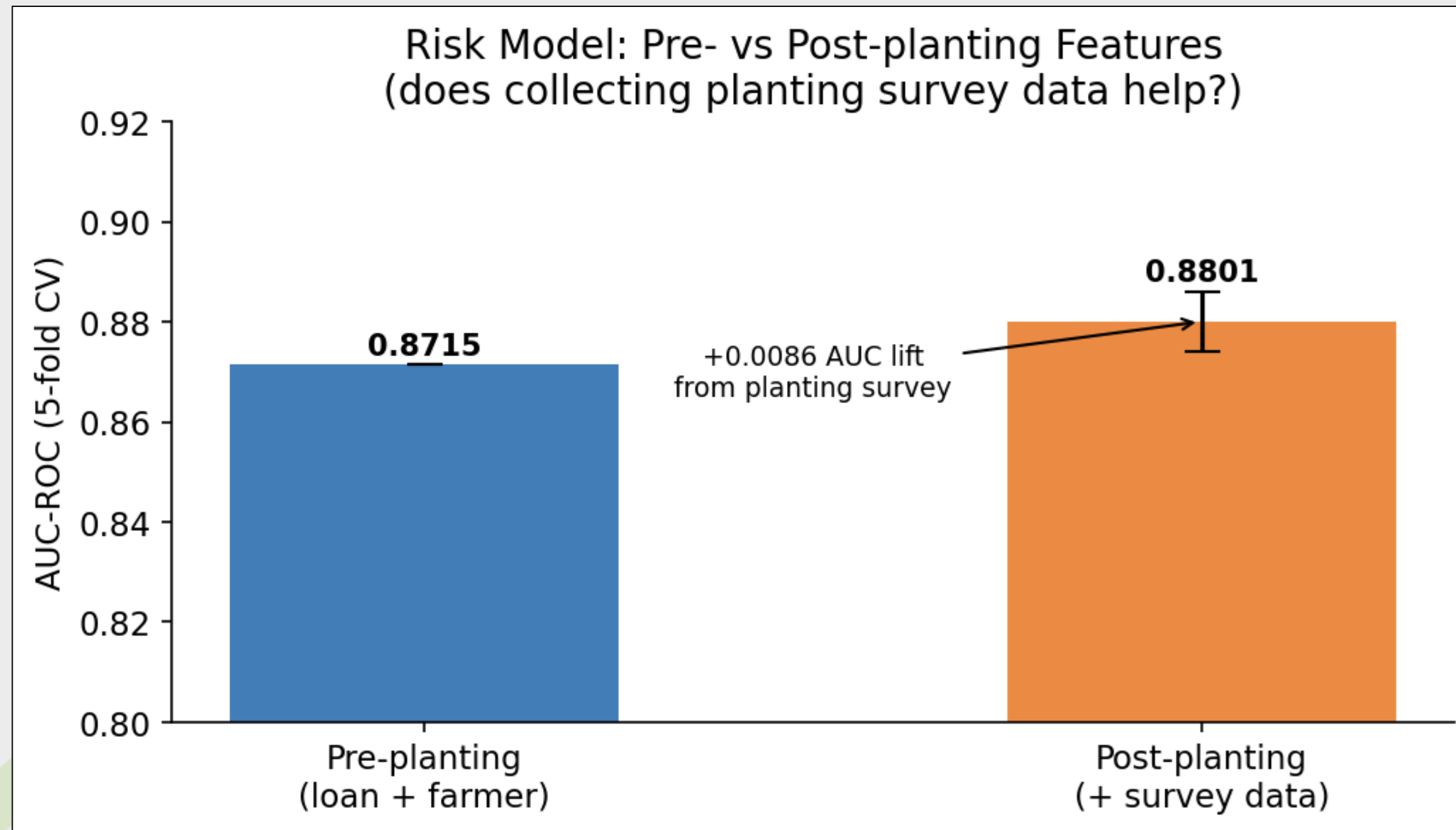


Business Insight

Fungicide must be made the priority intervention for HIGH-risk farmers and, the support for Partnership Programme farmers needs to be redesigned



Insight 9 : Risks must be flagged before the season starts



- **AUC 0.871** using only loan registration data — before a seed is planted
- Full model (+ planting survey) reaches **AUC 0.880** (+0.009 lift)
- Risk can be flagged at the moment of loan approval, months before season begins
- Enables proactive support, not reactive damage control

5254

FARMERS CAN BE RECOVERED!



Actionable Recommendations

1. Make fungicide the default in every Soy Bean package

45% of Soy Bean farmers — Good Nature Agro's largest crop have no fungicide. Those with fungicide yield 464 kg vs 160 kg without, a 189% difference. This is the single highest-return input decision available.

1. Deploy the risk model at loan registration

The model flags 27% of farmers and catches 74% of non-sellers using only data that exists before a seed is planted. Apply HIGH-risk flags to trigger field visits, reduced loan exposure, and targeted early support.

1. Invest disproportionately in Season 1 farmer retention

Season 1 is where Good Nature Agro loses the most farmers (27.7% dropout). Farmers who survive it drop to 9.6% by Season 2 and 1.3% by Season 6+. Every Season 1 farmer retained is a long-term asset. This is the intervention with the highest lifetime return.



Conclusion



Our model built entirely from existing data can identify 3 in 4 at-risk farmers before a seed is planted.

The 5 recommendations do not require new data collection, new infrastructure, or new programmes. They require applying our models!